



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-2026

Semester: I

Examination: ISE II

Class: T.Y. (CSE)

Branch: CSE

Date: 11/10/2025

Time: 11:00 To 11:45

Subject: Design and Analysis of Algorithms

Code: PCC-CS305

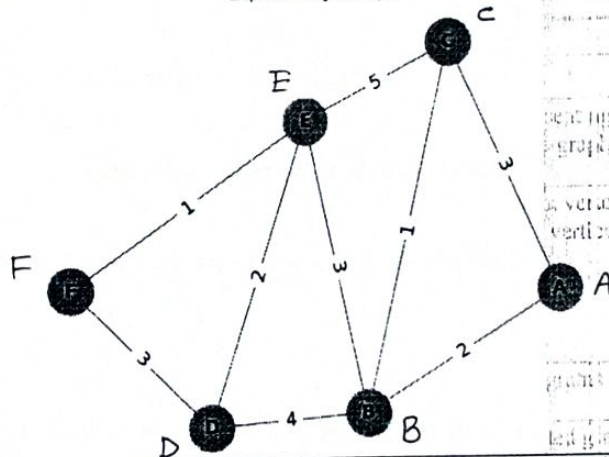
Max. Marks: 20

Solve All Questions

Marks

- Q.1.** Differentiate between an Adjacency Matrix and an Adjacency List for representing a graph. Mention the complexity of each in terms of V (vertices) and E (edges). Assume a suitable graph $G(V,E)$ and represent it using Adjacency Matrix and an Adjacency List. (5)
- Q.2.** For the given directed graph, perform a Depth First Search (DFS) starting from vertex A. List the vertices in the order they are visited (discovery order). When there's a choice, visit the vertices in alphabetical order. (5)
- Vertices: A, B, C, D, E, F
 - Edges: (A, B), (A, C), (B, D), (B, E), (E, F)
- What is the time and space complexity of DFS algorithm?
- Q.3.** Explain Spanning Tree and Minimum Spanning Tree (MST). Use suitable diagrams for explanation. What are the applications of MST? (5)
- Q.4.** Find the Minimum Spanning Tree (MST) for the following weighted, undirected graph using either Prim's or Kruskal's algorithm. Show the step-by-step selection of edges and calculate the total weight of the MST. Also compare Prim's and Kruskal's Algorithm. (5)

Graph with Weighted Edges



Course Outcomes (COs):

- CO1: Analyze the efficiency of algorithms by determining their time and space complexities.
- CO2: Apply Divide and Conquer, Greedy, Dynamic Programming, and Backtracking techniques to solve complex problems effectively.
- CO3: Implement algorithms to solve optimization problems like Knapsack, Job Sequencing, Matrix Chain Multiplication, and Traveling Salesman Problem.
- CO4: Design and analyze graph algorithms for shortest path, spanning tree, connectivity, and Eulerian paths.
- CO5: Differentiate between P, NP, NP-Hard, and NP-Complete problems, and demonstrate understanding through appropriate examples.
- CO6: Apply appropriate algorithmic techniques to design optimal or near-optimal solutions for computationally challenging problems.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6- Create

Question No	CO No	BT No
1	1, 4	1, 2, 3, 4
2	1, 2, 4	1, 2, 3
3	4	1, 2
4	4	1, 2, 3, 4



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year : 2025-26 (Semester I) Class : T. Y. B. Tech (CSE)
In Semester Evaluation – II Examination
Subject : Operating Systems Code : PCC-CS304/CS303
Date: 9/OCTO/2025 Time: 11:00 to 11:45 Max. Marks :20

- Note:**
1. Please read the questions **carefully**.
 2. Marks are shown at the end of each question on the right hand side .
 3. Marks will be deducted if answers are not written in **Neat handwriting** .
 - 4). Do not write **essay**.

1). What are the Latencies in the kernel ? [4]

Explain clearly along with appropriate diagram how to overcome Priority Inversion . [3]

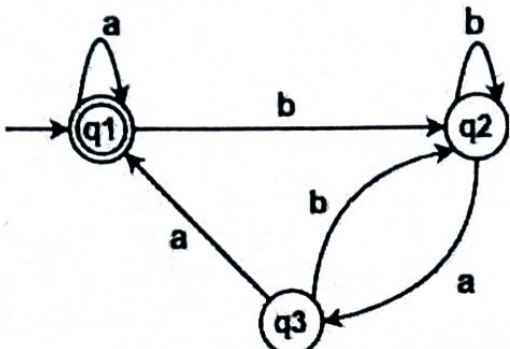
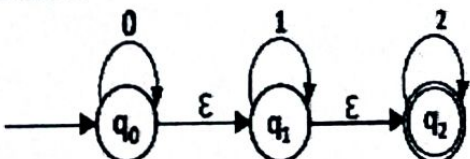
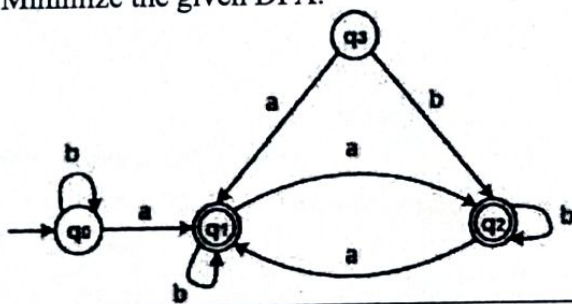
Explain different parameters characterizing the behavior of schedulers [3]

2). a) Draw the diagram a Modern Scheduler. Explain clearly its working. [5]

b) Draw and explain the Memory Layout of a process with and without threads [5]



**SHRI GURU GOBIND SINGHJI INSTITUTE OF ENGINEERING &
TECHNOLOGY, VISHNUPURI, NANDED**
Department Of Computer Science & Engineering
T. Y. B. Tech(CSE)
ISE II (2025-2026)

Date: 10 th Oct. 2025	Time: -11:00-11:45	Subject: -Formal Language and Automata Theory (PCC-CS303)(PCC-CS301)	Marks-20
Q.No.	Questions	Marks	CO(BT)
Q.1	Construct the regular expression for the given finite automata (FA) using Arden's Algorithm. 	04	CO2(BT3)
Q.2	Convert the following ϵ -non-deterministic finite automata (ϵ -NFA) to its equivalent deterministic finite automata (DFA) by using the ϵ -closure method. 	04	CO1(BT2)
Q.3	Minimize the given DFA. 	04	CO1(BT3)
Q.4	Design an equivalent DFA from the given regular expression: $10+(0+11)0^*1$.	08	CO2(BT3)

***** All the Best *****

CO's	Statement
CO1	Understand the fundamental concepts of automata and formal languages.
CO2	Analyze and construct finite automata (FA) and regular expressions (RE).
CO3	Demonstrate knowledge of context-free grammars (CFGs) and context-free languages (CFLs).
CO4	Examine pushdown automata (PDA) and their relationship with context-free grammars.
CO5	Understand how Turing machines (TM) and computational theory work
CO6	Explore the applications and limitations of formal language theory.
BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create	